

National University of Computer and Emerging Sciences, Chiniot-Faisalabad Campus



Course Name:	Applications Information and Communication Technology	Course Code:	CL1001
Degree Program:	BS(CS) BS(SE) BS(AI)	Semester:	Fall 2023
Exam Duration:	130 minutes	Total Marks:	75
Paper Date:	Monday, January 1 st , 2024	Obtained Marks	
Sections:	ALL	No of Page(s):	3
Exam Term & Type:	Final Term I Closed Book	Required Answer Book:	Yes
Course Instructor	Ms. Kashaf Shakoor, Ms. Nabeela Ashraf, Ms Areeba Waseem, Ms. Sahar Ajmal, Mr. Muiz Qadir, Ms. Aqsa		

Student: Name: _____

Instruction/Notes: _____

Roll: _____

Section: _____

Questions. Calculators are allowed.

Q:3 5 Marks	Q:4 15 Marks	Q:5 15 Marks	Q:6 15 Marks	Q:7 15 Marks	Q:8 10 Marks

Q:3

- i. Draw Computer architecture diagram and provide definition of each component? /5 CLO-1

Q:4. Attempt all parts. /15 CLO-2

- Calculate the 2's complement of the binary number $(1101101011)^2$. /2.5
- Convert the result from part (1) into its hexadecimal equivalent. /2.5
- Multiply the binary number $(1101101011)^2$ by the octal number $(37)_8$. /2.5
- Convert the product obtained in part (3) into decimal. /2.5
- Take the 1's complement of the binary number $(101010110011)^2$. /2.5
- Divide the binary number from part (5) by the hexadecimal number $(A3)^{16}$. /2.5

Q:5.

- Write a note on types of transmission modes in computer networks. /6 marks CLO-5
- Write differences between the user and kernel modes in operating systems. /5 marks CLO-3
- Define multiprogramming operating system and context switching. /4 CLO-3

Q:6

a. You have given 5 entries of attendance record. Your task is just to write the formulas for /10 marks CLO-3

- Find total present students /2
- Find total absent students /2
- Find percentage of present records. $(TotalPresent * 100 / totalClasses)$ /2
- If percent of present is less than 80 then show D-Bar, otherwise Regular /4

Note: You can find the above points (1-4) for just only one student. i.e. find all the above results just for Ali or any other student mentioned below

	A	B	C	D	E	F	G	H	I	J
	Names	Class1	Class2	Class3	Class4	Class5	Total Present	Total Absent	Total Present Percentages	D-Bar/Regular
1										
2	Ali	1	0	1	1	1				
3	Ayesha	1	1	0	0	0				
4	Aimen	1	1	1	1	1				
5	Usman	1	0	0	1	1				
6	Adil	1	0	0	1	0				
7	1 - Present									
8	0 - Absent									

Sample excel file

	A	B	C	D	E	F	G	H	I	J
	Names	Class1	Class2	Class3	Class4	Class5	Total Present	Total Absent	Total Present Percentages	D-Bar/Regular
1										
2	Ali	1	0	1	1	1	4	1	80	Regular
3	Ayesha	1	1	0	0	0	2	3	40	D-Bar
4	Aimen	1	1	1	1	1	5	0	100	Regular
5	Usman	1	0	0	1	1	3	2	60	D-Bar
6	Adil	1	0	0	1	0	2	3	40	D-Bar
7	1 - Present									
8	0 - Absent									

Solved file

b. Write the basic view(s) in MS-Access. Just write their names. /2

c. Write a note on the component(s) of Database System. /3

Q: 7

- In C++ you explicitly write data type with variables i.e int c=0; string str="this is string"; float flt=10.9;. But in JavaScript you don't explicitly write data types like in C++. So, what is the method to write data types for JavaScript? Justify your answer. /5 CLO-4
- How HTML DOM is a standard object model and programming interface for HTML? Justify your answer by giving 5 points. /5 CLO-4
- When anything in JavaScript is string it will perform operators with respect to string type i.e. addition on string will perform append operation instead of addition. What is the solution for this error? Justify your answer with code. /5 CLO-4

Q:8

- Answer the following questions using the concepts of the HTML/CSS/JS /10 CLO-4
 - Explain the purpose of the following HTML tags: /1
 - <head>
 -

 -
 - <form>
 - Create a basic HTML structure for a webpage that includes a title, a heading, a paragraph, a Link, and an image. /2
 - Write a CSS rule to style a class named "highlight" with a yellow background color, bold text, a font size of 16px and 30px padding. /2

iv. Write a code for adding two field values using JavaScript? /3

Value 1:

Value 2:

Result:

v. Create a button that, when clicked, changes the background color of a specific element on the page. Use both HTML and CSS for this task. /2